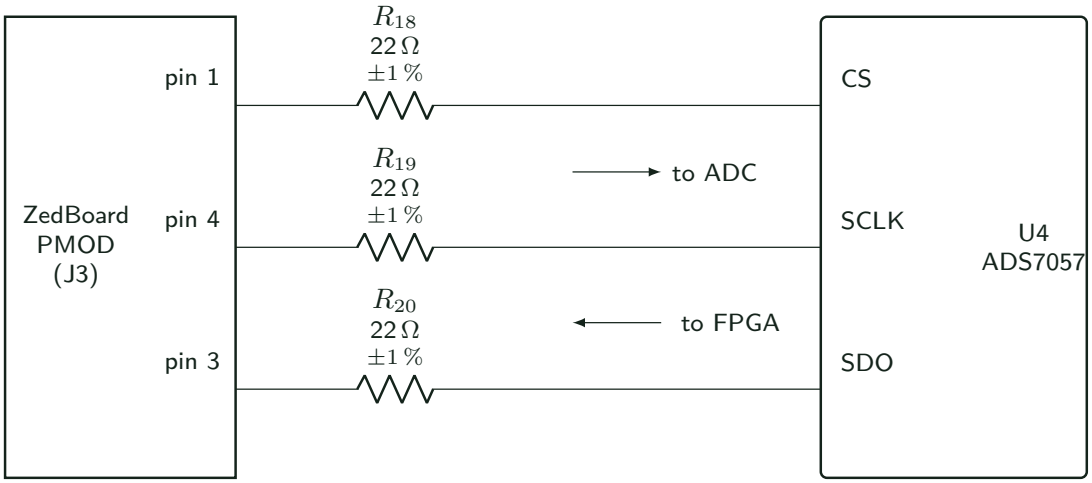
Slide 9: the SPI link

The protocol to the ZedBoard

ADS7057 test board v3 — study notes

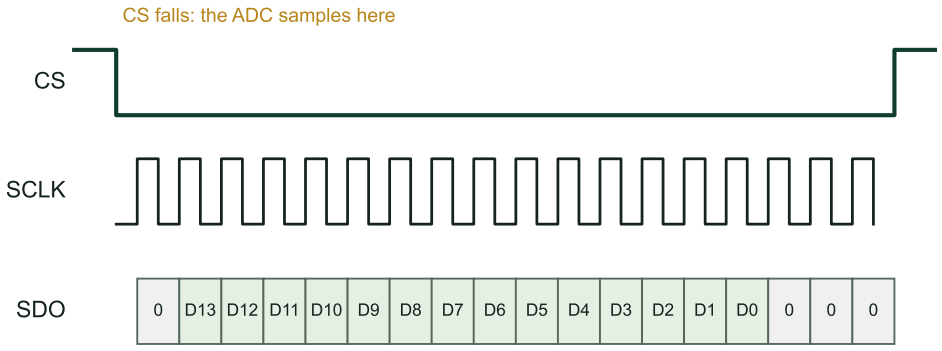
1 The wires



PMOD pins 5, 11 GND; 6, 12 3.3 V; 7 BIG_N; 10 WAKE_N; 2, 8, 9 not used

Three signal wires (CS, SCLK, SDO) and the two listener lines. There is no MOSI because the ADC only talks. The 22 Ω resistors sit at the ADC. With about 10 pF of trace and pin (estimate), $\tau = 22\,\Omega \times 10\,\text{pF} = 0.22\,\text{ns}$ against a bit time of 16.7 ns at 60 MHz. They damp ringing and cost nothing in speed.

2 One frame



18 clocks: leading 0, 14 data bits (MSB first), 3 zeros. After the 18th clock CS rises; at least 95 ns of acquisition follow.

CS falls and the ADC samples at that moment. The host sends 18 clocks. SDO shows a leading 0, the 14 data bits (MSB first) and three zeros. After the 18th clock the ADC converts nothing more. CS rises and the ADC acquires the next sample for at least 95 ns.

3 The data is one frame late

The ADC sends the result of the previous conversion while it converts the current one. Frame 1 returns zeros, frame 2 returns sample 1, frame 3 returns sample 2. The FPGA must tag each word with the time of the *previous* CS fall.

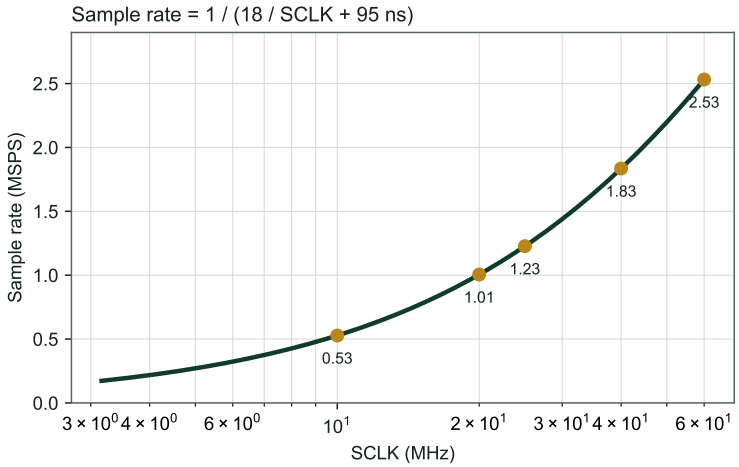
4 Power-up

The datasheet recommends an offset calibration. The first frame has 24 clocks, not 18. The ADC disconnects its input, corrects its offset and sends zeros. At 20 MHz this takes $24/20\,\text{MHz} + 95\,\text{ns} = 1295\,\text{ns}$. Repeat it when AVDD or the temperature changes a lot.

5 How fast can it go

One sample costs 18 clock periods plus the acquisition time:

$$T_s = \frac{18}{f_{\text{SCLK}}} + 95\,\text{ns}, \qquad F_s = \frac{1}{T_s}.$$



SCLK	one sample	rate
10 MHz	1895 ns	0.53 MSPS
20 MHz	995 ns	1.01 MSPS
25 MHz	815 ns	1.23 MSPS
40 MHz	545 ns	1.83 MSPS
60 MHz	395 ns	2.50 MSPS

The FPGA drives the ADC pins directly at 3.3 V (no level shifters on this board). Start at 10 to 20 MHz and raise the clock after the first test. The sample rate sets the fold frequency of the anti-aliasing filter (slide 6): 1.23 MSPS folds at 615 kHz.

On the 16-channel backplane (docs: `doc/backplane_plan.md`) the link crosses level shifters, a ribbon and a buffer. The round trip adds about 20 ns (estimate). Sample SDO one clock later than a plain SPI master would.